5. Conclusions

The GCP seeks evidence for a subtle correlation between deviations in a distributed random system and human mental activity. The event experiment examines the normalized data from a synchronized global network of physical RNGs during periods of widespread collective human attention. We find that, while the data fluctuate near expectation over the 9-year extent of the database, the aggregate deviation of data during 236 registered *formal events* is significant at 4.5 standard deviations. The highly significant aggregate result rests on a rigorous protocol which determines all event parameters before the data are examined. The result is confirmed empirically by a re-sampling analysis on the full database. We have proposed the correspondence of data deviations with the identified events as an operational definition of global consciousness, and our analysis has shown this to be a productive approach. The present paper will serve as a general background and foundation for a series of detailed assessments of questions stimulated by these results.

The average effect is small and broadly distributed among the events. The small event effect size of 0.3 sigma has several consequences. Most notably, the effect is too weak to meaningfully interpret individual events. Single events are dominated by random noise and analytical tests require sets of 50 events or more to achieve a reasonable statistical power. Second, the small effect size permits a simple Gaussian model of the distribution of event z-scores. Modeling of the event distribution suggests that roughly two-thirds of the event outcomes correspond to true positive deviations. The model finds that the remainder of the distribution contains both negative and null deviation events. One implication is that our standard prediction of positive variance deviations is not sufficiently sophisticated. Preliminary work addressing the effects from distinct categories of events included in the database reveals substantial differences, and may lead to the identification of categories that tend to produce negative versus positive deviations, or to produce null effects. Third, the existence of true negative or null events implies that the measured effect size estimates a lower bound on the effect size of positive deviation events.

A thorough analysis of the statistics used in the formal experiment shows that the effect is due to deviations in the variance of the network at 1-second resolution. The variance excursions are driven by correlations between or among the geographically separated RNGs in the network and are not due to changes of the output distributions of the individual RNGs. That is, we find no evidence for shifts of the mean or changes in the variance of the devices. This result stands in contrast to studies of RNG deviations in intending-subject RNG experiments which typically find shifts in the device mean. It also differs from the device variance excursions found in field research using a single RNG to study group consciousness. The GCP result thus suggests an entirely new phenomenon comprising correlations among globally distributed systems in the mental and physical domains.

We remain cautious as to the interpretation of these results. Any possible extension of our operational definition of global consciousness to a *theoretical model* needs to be assessed against alternative explanations and a careful study of the data's statistical structure. The GCP hypothesis and the physical implementation of the project allow us to ask whether the geographical distributions of engaged populations and the RNG network play an essential role. It will be important to examine the data for dependencies on distance and network density, particularly in light of the finding that the variance deviations arise from RNG-RNG correlations. Indeed, a complete absence of distance structure would obviate any need for a globally deployed network. Other structural parameters to investigate include time – the effect of varying the size and placement of the time window defining the event; the relative contributions of individual network nodes; and higher order inter-node correlation statistics. Evidence of structure in these or other parameters would indicate that the network is exhibiting a more complex behavior than is indicated by the simple deviations we have measured. To the extent data structure is found, models need to accommodate the possibility that the network is producing true data anomalies.

An alternative explanation is that psi intuition on the part of the experimenters may result in a fortuitous choice of event parameters which favors an anomalously biased selection of the random data. While it is unlikely that experiments can exclude anomalous experimenter effects altogether, for such a selection model, the data would not be expected to have underlying structure. This provides a further motivation for examining the structure of the event data. Several preliminary analyses, which are beyond the scope of this paper, indicate that the question of data versus selection anomalies is amenable to analysis. This preliminary work suggests that data anomalies may indeed be present in the event data. We find that the RNG-RNG correlations driving the network variance decrease with inter-node distance, and we find that deviations in the event data show a characteristic decay if the specified event time is shifted. Structural features like these can be expected if the variance deviations depend on experimental parameters such as network geography and the time characteristics of events.

An objection can be made concerning the ubiquitous nature of collective mental activity and the apparently sparse occurrence of what we designate as global events (the events comprise less than 2% of the database). If there is a global consciousness effect, why is it not evident throughout the database? A possibility is that measurable data deviations only correlate with those events which engage the largest numbers of people or evoke the strongest emotional responses. One way to test this idea is by devising a binary classification of event magnitude. A qualitative division of events into sets of major and minor magnitude is reasonably straightforward, given the wide range of events in the registry (ranging from the September 2001 terrorist bombings to Pierre Trudeau's funeral). We find that the subset of major events has greater network variance deviation, with the difference from the minor event set significant at the 0.05 probability level. An explanation for the lack of anomalous deviation in the full database may then rest on the degree of coherence of worldwide mental activity, which might be appreciable only for truly large events. Simultaneous instances of separate collective activities, which are not mutually connected or coherent, may correspond to data with essentially null correlation structure.

This paper is intended to succinctly document the GCP event experiment. We have shown that the data are indeed random, with parameters indistinguishable from theoretical predictions when examined as a whole. However, the data segments corresponding to the global events specified in the formal experiment do differ from expectation with high statistical significance, as predicted in the statement of the experimental hypothesis. The deviations we measure are associated with RNG correlations which extend over global distances. This result suggests a subtle and far-reaching interdependence between mind and matter. The implications of these findings for both physical and psychological models seem profound, but much work needs to be done to illuminate the issues. We will proceed with focused analyses addressing questions designed to deepen our understanding of the GCP results.

Acknowledgments

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Appendix 1

The wide variety of events included in the formal experiment is a reflection of the GCP's exploratory character. The criterion for event designation, as stated in the experimental hypothesis, is an instance of collective attentional response among people separated by global distances. In order to fully accommodate this

TABLE A1.1
A Sample of the Formal Events Chosen to Test the GCP Hypothesis

Number	Event description	Begin date time	End date time
1	US Embassy bombings Africa	1998-08-07 07:15:00	1998-08-07 10:14:59
21	Earthquake in Colombia	1999-01-25 17:15:00	1999-01-25 21:14:59
43	New Year variance 2000	1999-12-31 09:30:00	2000-01-01 11:29:59
56	Pierre Trudeau funeral	2000-10-03 15:00:00	2000-10-03 16:59:59
80	Terrorist attacks Sept. 11 2001	2001-09-11 12:35:00	2001-09-11 16:44:59
90	World-wide meditations	2001-11-11 11:00:00	2001-11-11 11:14:59
121	Wellstone plane crash	2002-10-25 15:00:00	2002-10-25 16:59:59
122	Chechen hostage tragedy	2002-10-26 02:30:00	2002-10-26 04:59:59
131	Global peace demonstrations	2003-02-15 00:00:00	2003-02-15 23:59:59
180	Athens Olympic opening	2004-08-13 18:00:00	2004-08-13 20:59:59
181	Day of murderous violence	2004-08-31 00:00:00	2004-08-31 23:59:59
182	Republican Convention Bush	2004-09-03 02:09:59	2004-09-03 03:11:59
183	Russian school hostages	2004-09-03 05:00:00	2004-09-03 08:59:59
184	Earthdance, 2004	2004-09-18 22:50:00	2004-09-18 23:14:59
197	Pope John Paul's funeral	2005-04-08 08:00:00	2005-04-08 12:29:59
223	TM flyer aggregation	2006-07-29 12:30:00	2006-09-09 23:29:59
255	Benazir Bhutto assassination	2007-12-27 11:00:00	2007-12-27 18:59:59
259	Attacks in Gaza	2008-03-01 00:00:00	2008-03-01 23:59:59

TM = Transcendental meditation.

criterion, it is necessary to investigate a range of event types. Accordingly, selected events may engage large or small populations, be tragic or celebratory, or depend on either natural or human circumstances. It is evident that with this approach we may identify some event types that do not correspond to periods of global consciousness. However, this possibility underscores the importance of employing an inclusive event selection procedure, as it permits a thorough exploration of the experiment's hypothesis. Table A1.1 shows a selection of *formal events* drawn from the prediction registry.

Appendix 2

The modeling procedure for estimates of positive, negative and null event fractions uses the sum of three standard normal Gaussian distributions with positive, zero and negative means, respectively. Because the effect size is small and there are no outliers, the variance of each Gaussian is dominated by the null distribution and the model variances can be set to one. A more elaborate approach would accommodate a range of mean values for both the positive and negative deviating fractions, but simulations show that this does not improve the model. This is due to the no-outlier/small-effect condition which constrains the range of positive (negative) model means so that a single Gaussian for each deviation direction is sufficient. The model can be expressed as follows:

$$G_{\text{model}}(A, B, C, \mu_{+}\mu_{-}) = \sum Ag_0 + Bg_{\mu+} + Cg_{\mu} \qquad \text{(A1.1)}$$

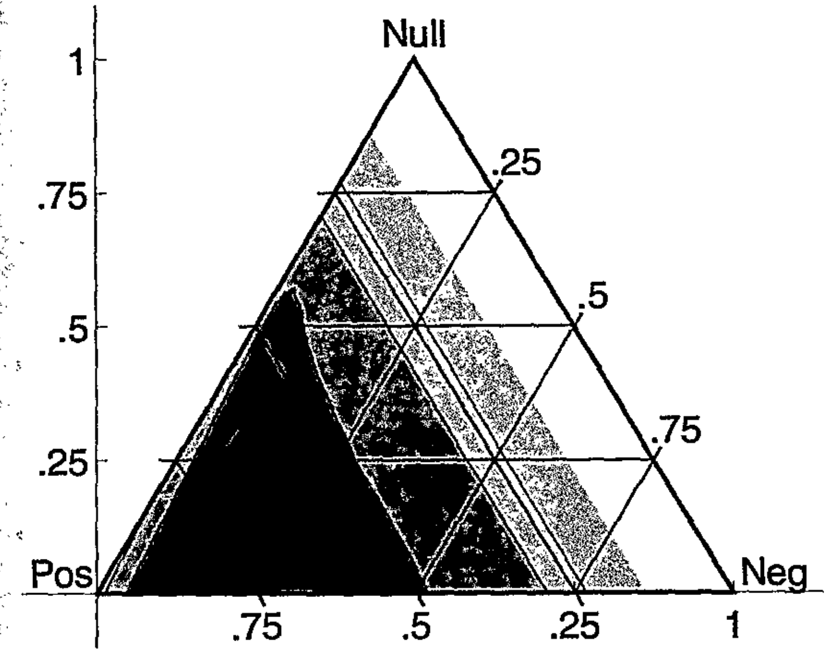

Fig. A2.1. The composition fractions for best fits of a ternary model of positive, null and negative deviation event z-scores. The model finds that the z-score distribution is best described by a large fraction of positive events and potentially non-zero fractions of both null and negative deviation events. The contours represent fit probabilities in 5% steps, with the maximum contour of 20% indicated by the darkest shading.

where g is a Gaussian distribution function. The model parameters are the fraction coefficients $\{A,B,C\}$, which are constrained by $A+B+C=1$, and the distribution means of the positive and negative fractions, $\{\mu_+,\mu_-\}$.

Goodness-of-fit tests provide a map of the "compositions" of positive, null and negative-going event scores compatible with the data, for a range of $\{\mu_+,\mu_-\}$. We have examined the model over the full span of fraction coefficients and for a wide range of mean values. For all of the 9800 models tested, the experimental z-scores are binned into a unique set of 14 bins which are selected to yield a model expectation >5 for all bins. A model goodness-of-fit is determined as a chi-squared probability of the mean-squared error, on 12 degrees of freedom (the composition constraint and an amplitude factor reduce the *df* by 2). A low fit probability indicates that random measurement error accounts poorly for the fit error and is grounds for rejecting such a model. Since we are interested in the composition fractions, we project the five-parameter results from the 9800 models into composition space. The projection can be represented in a ternary composition diagram as shown in Figure A2.1. The vertices of the triangular diagram are points of pure unitary composition for the positive, negative and null fractions, as labeled in Figure A2.1. Parallel grids are *lines of constant composition* for the fraction-type facing the grid lines. For example, the horizontal grids are lines of constant null fraction. The shaded contours are lines of constant fit probability, in increasing steps of 5%, from lightest to darkest. The dark contour in the lower left corner, which is the region of best fit, delineates a region with fit probability >20%, indicating that the model gives an accurate representation of the data for compositions within the contour.

The model yields a region of preferred fractional composition which includes potential contributions from both null and negative-going events. Whereas the best fit contour includes in its range a null fraction of zero along the horizontal axis, the fit probabilities decrease sharply as the line of zero negative-going composition (the left edge of the diagram) is approached. This suggests that the experimental z-score distribution does include a minority fraction of negative variance events. As a conservative estimate of the event fractions we take the average of model parameters for fits with chi-square probabilities exceeding a 15% cutoff with positive event fractions >50%. This covers most of the lower left portion of the ternary diagram. For this region the positive event fraction is 67%. The fractions of null and negative deviation events are 16% and 17%, respectively (1-sigma uncertainties $\pm$10%). The corresponding average parameter values for $\{\mu_+,\mu_-\}$ are 0.56 ± 0.09 and -0.49 ± 0.20. These are substantially larger in magnitude than the average event z-score of 0.30.

As with any analysis based on modeling, the z-score distribution model should be interpreted with caution. The parameter averages are derived from binned fits of a limited number of events and there is some sensitivity to tail occupation. Nevertheless, the average model parameters are robust against changes in bin selection, the probability cutoff level and small alterations in the z-score tail distributions. The model provides support for the reasonable supposition that the mean event z-score is a lower bound to the event effect size. It emphasizes that the anomalous variance deviations may be negative for some events.

Appendix 3

We can verify the significance of the 1-second network variance empirically by a re-sampling analysis. The interpretation of the result for 212 events given in the text, $Z_{Tot} = 4.10$, assumes that Z_{Tot} is effectively drawn from a standard normal distribution and that its significance thus can be represented by the probability value for a normal z-score of 4.10. The careful preparatory vetting and normalization procedures for all RNG trials, combined with the finding that the trial values for the event data conform to normality, supports this interpretation. The logic here is that Z_{Tot} should distribute normally, under the null hypothesis, because the underlying data trials distribute normally. However, it is worthwhile to estimate the significance of Z_{Tot} without relying on these assumptions. An empirical distribution for Z_{Tot} can be constructed by randomly selecting data periods corresponding to the events from the full database and calculating a set of $\{Z_{Tot}\}$ from the sampled event z-scores, Z_E. The significance of the value $Z_{Tot} = 4.10$ for the true event set can then be estimated from the empirical distribution, $\{Z_{Tot}\}$.

The randomly sampled Z_E are derived from solving $\mathrm{CDF}[\chi^2] = \mathrm{CDF}[Z_E]$ for Z_E, where CDF is the cumulative distribution function of the chi-squared and standard normal distributions, respectively. χ^2 is the re-sampled network variance for the event. As described in the text, the event set uses 212 events, instead of the full 236. We remove 19 New Year's events which are not easily adapted to the network and device variance recipes, as well as 5 events which have extremely long durations. The removals are inconsequential since we are merely comparing recipes and testing our approximations.

The re-sampling procedure is done with replacement. We calculate the Z_E for all of the 212 events by selecting random start times for each event from within the entire 9-year database. The Z_{Tot} is then calculated and the procedure is repeated 100,000 times. The resulting distribution has just one Z_{Tot} exceeding the value of 4.10, which gives an empirical probability value for the experimental Z_{Tot} of 10^{-5}, or an effective Z = 4.26. A more accurate determination of the p-value requires populating the far tails of the distribution, for which roughly a million iterations are needed, which is at the limit of our computational capacities. As an alternative estimate, the distribution of 100,000 values is tested for standard normality, which obtains, and the experimental value of 4.10 is corrected by the mean of the empirical distribution. The re-sampling distribution mean is –0.23, which is consistent with an apparent negative trend in the database. The empirical estimate of Z_{Tot} calculated by this method is $Z = 4.33$, close to the direct empirical value of 4.26. These estimates give a slight *increase* in the significance of the experimental result relative to the value based on the theoretical chi-squared statistics.

A second approach is to remove the trends of the chi-squared statistics locally about each event and then to use the trend-subtracted values to calculate Z_{Tot} from the formal event periods. The trends are estimated by smoothing the database of chi-squared statistics with a Gaussian window ±7 days about each datum in the event. The window size is chosen to be much larger than the maximum event length of 1 day, but local enough to compensate for trends about the event period. This procedure yields a value of $Z_{Tot} = 4.29$, in close agreement with the re-sampling analysis.

Notes

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[2] Radin, D. I., & Nelson, R. D. (1989). Evidence for consciousness-related

anomalies in random physical systems. *Foundations of Physics, 19*, 1499–1514.

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[12] Radin, D. I. (2004). Electrodermal presentiments of future emotions. *Journal of Scientific Exploration, 18*, 253–274.

[13] Utts, J. M. (1991). Replication and meta-analysis in parapsychology. *Statistical Science, 6*, 363–403.

[14] Each registry entry is a prediction of a specific response by the network to an event. This should not be confused with predicting that an event will occur. The entries are made before data are examined, prior to the event if it is known, but afterward for events that are unpredictable, such as terrorist attacks, aviation accidents, or earthquakes. See Appendix 1 for examples showing the range of formal events.

[15] Nelson, R. D. (2001). Correlation of global events with REG data: An Internet-based, nonlocal anomalies experiment. *Journal of Parapsychology, 65*, 247–271.

[16] Nelson, R. D., Radin, D. I., Shoup, R., & Bancel, P. A. (2002). Correlations of continuous random data with major world events. *Foundations of Physics Letters, 15*, 537–550.

[17] The Orion RNG is available at http://www.randomnumbergenerator.nl/. The distributor for the Mindsong REG, Mindsong, Inc., is no longer in business. See http://noosphere.princeton.edu/reg.html. Three PEAR devices were custom built for the Princeton Engineering Anomalies Research program.

[18] Jahn, R. G., Dunne, B. J., & Nelson, R. D. (1987). Engineering anomalies research. *Journal of Scientific Exploration, 1*, 21–50.

[19] Public access to GCP database via the Custom Basket Data Request form at http://noosphere.princeton.edu/data/extract.html.

[20] The data trials are standardized binomial variables. The binomial distribution differs from the normal distribution in its fourth moment and the binomial (200, ½) has a theoretical kurtosis of 2.99, as opposed to 3 for the normal distribution. The kurtosis is not modified by the standardization procedure.

[21] Of 257 events in the prediction registry, 21 are excluded from the formal experiment for technical or methodological reasons. Network instabilities resulted in frequent null trials and potential biases during the first weeks of data acquisition. The network achieved stability in December 1998, and a re-sampling analysis has determined that event z-scores can not be reliably determined for the earlier data. Ten early events registered prior to stable operation have therefore been excluded. An independent review, commissioned by the project in 2002, identified 11 events with potential methodological errors or ambiguities. These events were excluded after the review was completed and an improved prediction registration procedure was adopted. An extensive re-examination of data integrity and prediction methodology completed prior to publication of this paper confirmed both the earlier event rejections and the acceptance of all registered events since 2003. Excluding the 21 rejected events from the experiment reduces the mean event z-score slightly. The value decreases from 0.33, the mean over all 257 registered events, to 0.30 for the 236 accepted events of the formal experiment.

[22] Nelson, R. D. (2006). *Anomalous Structure in GCP Data: A focus on New Year's Eve*. Proceedings of the Parapsychological Association Convention, Stockholm, Sweden, August.

[23] While the standard prediction is a positive deviation, the exploratory nature of the experiment provides an opportunity to identify event types or categories that may yield null or negative results. If we somehow knew that all events produced positive deviations, then the mean event z-score would be a valid estimate of the event effect size. However, since some events may be truly null or anomalously negative-going, the mean z-score estimates only a lower bound on the effect size magnitude.

[24] The autocorrelation formulation yields a more accurate estimate since it averages over all realizations of blocking (e.g., there are $N-1$ ways to initiate a blocking of block length N).